UNIT 5 – REVIEW

1. State in simplified form the derivative of the following:

   a) $y=\left(e^{\ln x}\right)^2$ b) $y=2^{\cos x}+\ln(3e)$ c) $y=\dfrac{\cos 2x}{\sin 2x\left(\cos^2 x-\sin^2 x\right)}$ d) $y=3^{\log_3(\cos 6x)}$

2. Differentiate each of the following. Simplify fully.

   (a) $y=3x^2\tan(2x)$ (b) $f(x)=\cos\left(x^3\right)$

   (c) $s=\sin(\ln 7t)$ (d) $y=e^x\ln x$

3. Differentiate the following and leave in simplified form.

   (a) $f(x)=\tan^4\left(3\ln x^2\right)$ (b) $f(x)=2\ln\left(\sqrt{1-x+x^2}\right)$ (c) $s(x)=e^{\pi}\ln(\sin x+\cos x)$

4. Find the equation of the tangent to $y=x\ln x$ which is parallel to $3x-y+7=0$.

5. Determine all points of inflection of $y=e^x\sin x$ on the interval $0\le x\le 2\pi$.

6. Determine the equation of the tangent to the curve $y=e^x+x$ which passes through the origin.

7. Determine any **point(s) of inflection** of the function $f(x)=2\sin^2 x-1$ on the interval $-\pi\le x\le\pi$.

8. A line with slope m passes through the origin and is tangent to $y=\ln\left(\dfrac{x}{3}\right)$. What is the value of m?

9. Find the values of a and b (a ≠ 0) such that f is differentiable everywhere

   (a) $f(x)=\begin{cases}\dfrac{\cos x}{\sin x+\cos x}, & x\le 0\\ ax+b, & x\ge 0\end{cases}$ (b) $f(x)=\begin{cases}\cos x, & x<\dfrac{\pi}{2}\\ ax+b, & x\ge\dfrac{\pi}{2}\end{cases}$

10. A rectangle has its vertices on the x-axis, the y-axis, the origin, and the graph of $y=-\ln x$ in the first quadrant. Find the maximum possible area for such a rectangle.

11. Show that for any values A and B, $y=Ae^{2x}+Be^{-4x}$ satisfies the equation $y''+2y'-8y=0$.